

Consent at the Speed of Bleeding

What it means to say yes to an oxygen carrier when the patient cannot say anything

Most writing about hemoglobin-based oxygen carriers asks whether they work. This post asks a prior question that gets far less attention: who agreed to receive them, and how?

The answer is uncomfortable, because the setting where a BHOC is most needed is the setting where consent is hardest. A person losing blood in a road accident, a battlefield, or a remote clinic is often unconscious, in shock, or drifting in and out of understanding. The product is experimental or restricted. The normal machinery of informed consent, which assumes a calm conversation with time to think, does not apply. What replaces it is a set of legal exceptions and moral compromises that most patients, and most advocates, have never examined.

The rule that lets a trial start without asking

In the United States, federal regulation allows a narrow exception to informed consent for emergency research. The rule, found in Title 21 of the Code of Federal Regulations at section 50.24, permits an investigation to proceed without the patient's consent when several strict conditions are met. The patient's life must be at risk. Available treatments must be unproven or unsatisfactory. Obtaining consent must be impossible because of the patient's condition, and there must be no time to reach a legal representative. The research must offer a real prospect of direct benefit. And, unusually, the sponsors must consult the community from which participants will be drawn, and publicly disclose the trial before it begins.

This is the framework under which the best-known trial of a hemoglobin carrier in trauma was run. The product was a human-derived polymerized hemoglobin called PolyHeme. Patients with severe hemorrhage were randomized in the field, in ambulances, to receive either the carrier or standard saline, and after arrival at hospital, either the carrier or red cells. Because the whole point was to test treatment in the first minutes, waiting for consent would have defeated the study.

The trial drew heated criticism from bioethicists, some of whom argued in print that community consultation had been thin, that the control arm meant some patients would receive an unapproved substance instead of blood available at the hospital, and that the risk information available at the time did not support an exception. Supporters replied that the exception existed precisely for cases like this, that public meetings had been held, and that people who did not want to participate could wear opt-out bracelets. Both sides described real things. Neither side was inventing.

What I want to highlight is a structural point. The exception treats consent as a property of a community rather than of an individual. Nobody in the ambulance said yes. A group of strangers in a city, at a meeting most residents did not attend, said yes on behalf of whoever might one day be put in that ambulance. That is a strange kind of permission, and it is worth understanding before anyone advocates for or against it.

The evidence that changed the terms

The moral weight of the exception depends on how big the risk is. That risk was recalculated in 2008, when a group of researchers published a pooled analysis of trials of several hemoglobin carriers across many indications. It combined sixteen trials and more than three thousand patients, and reported that treatment was associated with a higher risk of death and of heart attack than control. The absolute numbers were small in some categories, and the products were not interchangeable. But the signal was enough for the exception's basic premise, a prospect of direct benefit, to look shakier in hindsight.

It is important to be fair about what that analysis did and did not say. It pooled different molecules, made by different methods, at different doses, in different diseases. It did not show that every future product will cause harm. It showed that a class of first-generation products carried a hazard that individual trials had not been powerful enough to detect. From an ethics standpoint, this is the cleanest illustration of why the consent problem cannot be solved by quick permission. If the risk of a product is not known until years after a large trial ends, then a community that agreed to it in advance agreed to something nobody could describe.

Conscience: the patient who has already decided

Now turn to the reverse situation. There are people who can be quite sure they do not want a transfusion, and who are also in danger of bleeding to death. Jehovah's Witnesses, on the basis of their reading of scripture, refuse transfusion of whole blood, red cells, white cells, platelets, and plasma. They regard some fractions, such as albumin, as a matter of individual conscience. Many accept hemoglobin-derived products, but not all, and the decision differs from person to person.

For this community, a hemoglobin carrier has a moral profile that does not resemble that of an experimental trauma drug. It is an alternative to a treatment they will not accept. The question of consent is reversed: instead of "how do we permit a treatment the patient cannot approve," it becomes "how do we make sure the patient's refusal is honored, and how do we offer a substitute that does not violate it."

This is where the source of the hemoglobin matters. A bovine BHOC is not human blood. It contains no human cells or plasma. For many Witnesses, that distinction makes it acceptable, and it has been used in emergency cases where a patient with life-threatening anemia refused blood but agreed to it. In those cases the product was typically obtained by a single-patient emergency request, and the consent conversation took place with the patient, if awake, or through the documents the patient had carried.

That last phrase points to something practical. Witnesses commonly carry an advance medical directive, a card or form stating what they will and will not accept. In an emergency, that card is the consent. It is the one consistent place in medicine where a patient's answer exists before the question arrives. It also shows a gap: the directive usually lists blood components, and rarely mentions a bovine hemoglobin product. A patient who ticked "no fractions" might or might not have meant this. The clinician who finds the card cannot ask.

The single-patient request

Outside trials, the route to an unapproved product is expanded access, sometimes called compassionate use. A treating physician asks the manufacturer for the product and asks the regulator for permission, and in a true emergency the regulator can authorize by telephone. The paperwork that follows includes a consent form, an ethics committee review, and a report of what happened.

Two features make this route strange for oxygen carriers. The clock runs in hours, and a hospital that has never done it before must find a manufacturer, an ethics reviewer, and a pharmacist in the same afternoon. And the consent given is real only if the patient can give it. Many of the reported single-patient cases involved adults who were conscious enough to say yes, or whose relatives could speak for them. That is a much easier situation than the roadside. It also means the public record of these cases says little about the hardest scenario, the one that motivates most calls for access.

The disclosure nobody prints on the label

There is a second conscience issue that affects a much wider group. A bovine product comes from cattle. For a Hindu patient, cattle are sacred, and receiving a product made from cows may be a serious problem. For some observant Muslim and Jewish patients, the question turns on the animal's slaughter and the status of a derivative, and religious opinions differ. Vegetarians and vegans may object on ethical grounds. A porcine product would raise parallel concerns for other groups. A lugworm product would raise almost none of these, though nobody has seriously studied how patients feel about that.

None of this is exotic. Hospitals already handle it for animal-derived medicines such as heparin from pigs or gelatin capsules. The relevant lesson is that a BHOC consent form that says only "this product is derived from bovine hemoglobin" is not enough for the emergency setting, because the person who needs to know is not there to read it. If a product is going to be stocked in ambulances or forward field units, its source must be visible at a glance and its presence must be recorded in a way that follows the patient into the hospital.

Consent after the fact

Emergency medicine has a partial answer to the timing problem, and it is worth borrowing. When treatment was given without consent, the patient or their family should be told as soon as possible, and the patient should be able to withdraw from further participation and, where research is involved, from data use. This is called deferred or retrospective consent. It does not undo the treatment. It does restore the patient's standing as a person rather than a specimen.

I have noticed that discussion of deferred consent in the hemoglobin literature tends to be procedural: who sends the letter, on what day. The patient's actual experience is rarely described. Imagine waking up in a hospital bed, being told you were given a red solution made from cow blood, and being asked to sign a form. The person may feel grateful, or violated, or both. How a system handles that moment determines whether communities support the next trial.

What a decent system would look like

I would suggest five design principles that follow from the above.

Pre-consent should be encouraged where it is possible. People who anticipate risk, such as those who work in remote areas, who serve in the military, or who have conscience-based limits, should have a way to record their wishes about oxygen carriers specifically, not only about blood.

Labels and records should name the source. Species of origin, manufacturing route, and lot must travel with the patient.

Community consultation should be real. That means meetings with translation, with faith leaders, with groups who might object, and with people who have been through emergency care, not only with a committee.

Disclosure after treatment should be prompt and plain, with a clear route to ask questions and to opt out of research use.

Exception should not become habit. If a product has been used in enough patients to be described, then its risks can be explained in advance, and the emergency exception should give way to ordinary consent wherever a competent patient exists.

Why patient advocates should care

A foundation working on access to BHOC is usually asked for evidence of efficacy. It should also be prepared to answer questions on consent. Who gets to say yes? Who is protected when the answer cannot be given? How do people with conscience objections, religious or otherwise, find out what they are being offered?

An advocacy group that raises these questions first earns credibility. It signals that access is not being pursued at any cost. It also protects the patients who most need the product, because a trial that loses public trust will be closed, and the next one will be harder to open.

The oxygen carrier that saves a life at a roadside is a scientific accomplishment. The system that lets a stranger receive it is a moral one, and it has to be built with the same care.

Further reading

The regulatory text is 21 CFR 50.24. The pooled safety analysis of hemoglobin carriers was published in the Journal of the American Medical Association in 2008. Bioethics commentary on the PolyHeme trial appeared in several medical and ethics journals during 2006 and 2007. Guidance for clinicians treating Jehovah's Witnesses is published by hospital liaison committees and should be read alongside a patient's own directive.